Decode-Time Transition Ledgers for Diagnosing World-State Drift

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Abstract

Language-model agents often answer questions about evolving situations after many intermediate transitions, but final-answer accuracy alone does not reveal whether an error came from event extraction, state update, stale-fact invalidation, answer rendering, or a simpler prompt-format change. We study a train-free decode-time transition ledger that asks the evaluated backend to write candidate state updates, mark superseded facts, and reverify the answer against the resulting current state. The evaluation uses "gpt-5-mini" through an OpenAI-compatible backend on a completed diagnostic matrix with 2,211 scored rows across FANToM, MMToM-QA, and MuMA-ToM, eleven inference-time conditions, and 67 matched tasks per family-condition cell. The result is mixed rather than a method-win headline: ledger + reverify is slightly higher than direct answering on FANToM (60.4% versus 59.4%) but lower on MMToM-QA (68.7% versus 74.6%) and MuMA-ToM (77.6% versus 82.1%). Stronger controls also win in different families: FactTrack-like prompting leads FANToM, event reformatting leads MMToM-QA, and chain-of-thought leads MuMA-ToM. We therefore treat transition ledgers as an evidence-producing audit surface for world-state drift, not as a universal accuracy intervention, and we keep mechanism, oracle-gap, and resource-use claims weak where the completed artifacts do not support stronger conclusions.

Introduction

Models that interact with procedural text, stories, tools, or simulated users must maintain a current world state while older facts become stale. Recent work has begun to probe latent world states, time-aware fact tracking, and procedural world modeling (Feng, Russell, and Steinhardt 2024; Lyu et al. 2024; Li, Guo, and Andreas 2025; Chen et al. 2025), with older procedural and theory-of-mind benchmarks already showing that entity state and belief questions require more than static passage retrieval (Dalvi et al. 2018; Nematzadeh et al. 2018; Kim et al. 2023; Jin et al. 2024), while agent papers show that explicit traces, memory, and state prompts can alter behavior without changing model weights (Yao et al. 2022; Park et al. 2023; Xu et al. 2025; Rozanov and Rei 2025; Shinn et al. 2023; Madaan et al. 2023; Packer et al. 2023; Zhong et al. 2024). Benchmarks for theory-of-mind and procedural belief tracking further show that state questions can depend on perspective, event order, and entity-specific updates rather than on a single static passage representation (Feng, Russell, and Steinhardt 2024; Harang et al. 2025; Zhu et al. 2026; Xu et al. 2024; Chen et al. 2024; Wu et al. 2023). At the same time, recent theory-of-mind audits caution that benchmark performance can be brittle under small task changes and should not be read as a single general capability score (Ullman 2023; Kosinski 2024; Strachan et al. 2024; Riemer et al. 2024). The diagnostic problem remains: a final answer can be right for the wrong reason, and a wrong answer does not say whether the model missed a transition, kept an obsolete fact, or rendered the wrong entity.

This paper evaluates a transition-ledger intervention for that diagnostic problem. The intervention is deliberately train-free. For each episode, it asks the same evaluated backend to expose a sequence of state updates, invalidates older facts when later events supersede them, and then reverifies the answer against the ledger. The contribution is not a claim that such ledgers always improve accuracy. Instead, the contribution is a measured account of where this style of ledger helps, where it fails, and which stronger explanations remain unsupported.

The completed matrix covers FANToM, MMToM-QA, and MuMA-ToM with direct answering, chain-of-thought, ReAct-style scratchpads, StateAct-like state prompts, summary and reformatting controls, FactTrack-like prompting, ledger variants, and oracle/model ledger variants. The analysis supports three conservative claims. First, coverage is complete for the planned three-family by eleven-condition matrix. Second, the direction of the ledger effect reverses by family, so pooled headlines are misleading. Third, parsing failures are zero and therefore do not explain the reversal. It rejects the universal ledger-win claim and rejects the claim that ledger + reverify robustly beats prompt-format controls. Figure 1 summarizes this framing.

The study is designed as a diagnostic paper rather than a leaderboard paper. That distinction matters for world-state tracking because a method can create a more inspectable intermediate state while still lowering final-answer accuracy on some families. The evidence chain therefore records not only the result table, but also whether each intended claim is supported, weak, rejected, or missing. This keeps the narrative aligned with measured artifacts instead of allowing a successful-looking diagram to imply a stronger empirical result.

The paper's main positive contribution is the audit protocol. It shows how to execute a shared benchmark matrix, keep rows paired by task, evaluate multiple nontrivial inference-time controls, and translate those rows into conservative paper claims. The protocol is especially useful when a proposed controller has plausible mechanism stories but the measured family outcomes disagree. In that setting, a claim graph is not administrative overhead; it is the object that prevents a negative or mixed result from being overclaimed.

The empirical result is also useful because it identifies a failure mode for state-record prompting. Adding an explicit ledger can change task formatting and answer style at the same time that it changes state representation. The reformatted-events and chain-of-thought controls expose this confound. A ledger that beats direct answering in one family but loses to formatting or reasoning controls in another should be described as a diagnostic handle, not as a robust state-tracking solution.

Related Work

World-state tracking and procedural evaluation. Latent-state probes and state-tracking benchmarks ask whether models represent the changing variables needed for grounded prediction (Feng, Russell, and Steinhardt 2024; Li, Guo, and Andreas 2025; Zhang et al. 2025; Harang et al. 2025; Dalvi et al. 2018). Time-aware and procedural benchmarks such as FACTTRACK, FABLE, MET-Bench, and WorldPrediction stress entity attributes, temporal updates, and event-order reasoning (Lyu et al. 2024; Pallagani et al. 2025; Cohen and Mooney 2025; Chen et al. 2025). Theory-of-mind benchmarks add perspective-specific belief state as a parallel stress test for current-state maintenance (Nematzadeh et al. 2018; Kim et al. 2023; Jin et al. 2024; Xu et al. 2024; Chen et al. 2024; Wu et al. 2023). Our benchmark mix is narrower than a broad world-modeling suite, but it is aligned with the question this draft can support: whether an inference-time ledger exposes family-specific state-update behavior.

Agent memory and explicit state. Agent-memory systems and state-prompting methods motivate externalized state as a controller, not only as a hidden representation (Park et al. 2023; Xu et al. 2025; Hu et al. 2025; Rozanov and Rei 2025; Packer et al. 2023; Zhong et al. 2024). Broader agent scaffolds use reflection, self-feedback, tools, or API-oriented interfaces to change behavior at inference time (Shinn et al. 2023; Madaan et al. 2023; Schick et al. 2023; Qin et al. 2023; Li et al. 2023; Liu et al. 2023; Wang et al. 2023; Shen et al. 2023). PDDL-Mind similarly emphasizes belief reasoning with reliable state tracking (Zhu et al. 2026). These papers justify testing explicit state records, but they do not imply that any particular ledger format should win on every benchmark. Our claim graph therefore separates the existence of a completed matrix from stronger mechanism claims.

Inference-time reasoning baselines. Chain-of-thought and ReAct-style traces can improve reasoning by exposing intermediate steps or interleaving reasoning and acting (Wei et al. 2022; Yao et al. 2022; Wang et al. 2022; Yao et al. 2023). Verification and self-checking work motivates asking a model to inspect its own intermediate answer, but those methods also warn that verification prompts must be evaluated against strong controls rather than treated as mechanism evidence (Dhuliawala et al. 2024; Manakul, Liusie, and Gales 2023). This draft treats those methods as nontrivial inference-time baselines. The prompt-format audit is important because a ledger prompt changes more than state representation: it also changes answer format, event presentation, and the amount of text available to the model.

This related-work structure separates mechanism motivation from empirical support. Method papers need a concrete control loop before benchmark numbers can be interpreted, and agent-memory or reasoning-trace work should not be cited as automatic evidence for accuracy gains. These lines of work provide design pressure: external state, episodic memory, and scratchpad traces can make intermediate reasoning visible (Yao et al. 2022; Park et al. 2023; Xu et al. 2025), but each benchmark still decides whether that visibility helps the final task.

The closest conceptual neighbors are state-prompting and memory methods, yet the present protocol differs in what it claims. Memory and state-prompting methods often optimize future performance by retaining useful summaries, feedback, or explicit state descriptions (Rozanov and Rei 2025; Hu et al. 2025; Xu et al. 2025). Here the ledger is not optimized across episodes and is not trained. It is a within-episode record whose value is measured by paired comparisons and by whether it enables claim-level diagnosis. That difference is why the paper emphasizes rejected and weak claims alongside supported ones.

Diagnostic Setup

Each task contains an episode with events, entities, and a query about the current state. A direct condition asks for the answer without an explicit state ledger. A ledger condition asks the model to write transition records, invalidate superseded facts, and reverify the answer. Oracle and model-ledger variants separate manually supplied state structure from model-extracted state structure only as a diagnostic contrast; the completed results do not make this contrast a clean extraction upper bound.

We use four claim statuses. Supported claims are backed by the completed raw rows and deterministic derived artifacts. Rejected claims are contradicted by the measured matrix. Weak claims have some relevant rows but lack the labeled or statistical support required for a stronger statement. Missing claims identify evidence that was not produced in this stage. Table 3 gives the paper-facing version of that audit.

The family separation is part of the setup rather than an after-the-fact presentation choice. FANToM, MMToM-QA, and MuMA-ToM stress different forms of belief, perspective, and state update. A pooled average would mix task distributions whose best-performing conditions differ. Because the intervention can plausibly help on one family and hurt

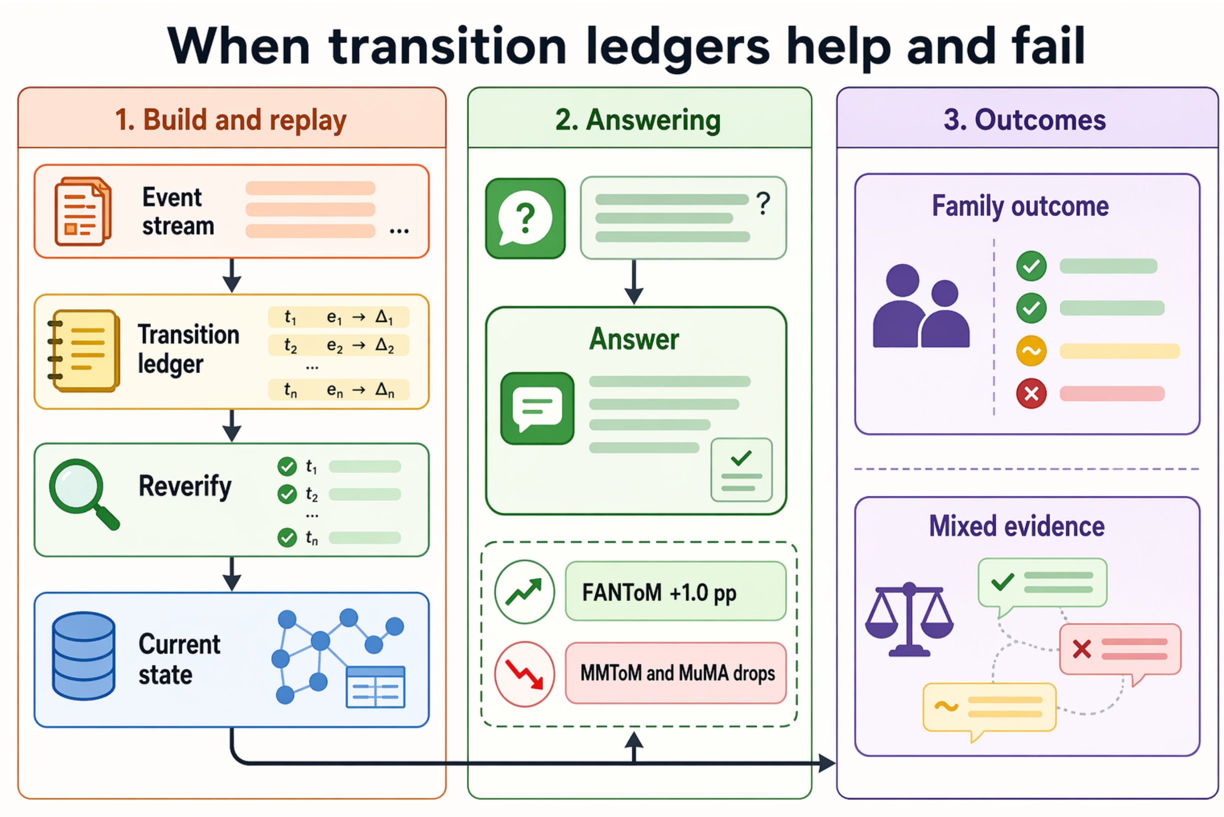

Figure 1: Teaser showing the central measured finding: ledger + reverify is diagnostic, not a universal winner, with a 1.0 pp FANToM delta and negative deltas on MMToM-QA (-6.0 pp) and MuMA-ToM (-4.5 pp).

on another, every headline sentence in the draft is required to preserve family-specific direction.

The diagnostic axes also separate extraction from use. A model-ledger row can fail because the transition record is wrong, because stale facts are not invalidated, or because the final answer ignores an otherwise correct record. An oracle-ledger row removes one source of extraction error, but the completed oracle/model contrast is not monotone across families. That is why the draft does not claim a clean extraction bottleneck.

Method: Transition-Ledger Diagnostic

The diagnostic pipeline in Figure 2 has five steps. First, the controller presents the benchmark episode and query under a condition-specific prompt. Second, ledger variants ask for state-transition records rather than only a final answer. Third, invalidation marks older facts as superseded when a later event updates the same state variable. Fourth, the answer is generated or reverified against the current state record. Fifth, a parser scores the answer and writes the result into the claim audit.

The method is intentionally conservative. It does not train a new model, tune a ranker, or change benchmark gold labels. It also does not treat the ledger text as intrinsically correct. The ledger is useful only insofar as it creates observable intermediate artifacts that can be paired with final-answer scores and ablations.

The evaluated paper system is a condition renderer, a hosted language-model backend, an answer parser, and a claim auditor. The rendered conditions are paper-facing prompt variants rather than learned policies. The evaluated model identifier recorded by the completed run is "gpt-5-mini" through an OpenAI-compatible API backend. The recorded evidence does not establish a separate temperature or decoding sweep, so this draft treats the run as one fixed backend-and-prompt protocol and does not claim hyperparameter robustness. Resource accounting is retained as replay metadata rather than interpreted as a primary scientific claim.

The pipeline has no hidden training stage. The same episode is rendered under condition-specific instructions, and the answer parser applies the same scoring contract across conditions. Ledger rows add text and structure, so the fair comparison is not simply ledger versus direct. It also includes chain-of-thought, ReAct-style scratchpads, StateAct-like state prompting, event reformatting, summarization, FactTrack-like prompting, and no-invalidation variants. These controls make it possible to distinguish a state-record effect from a prompt format effect.

The invalidation step is intentionally simple: later updates can supersede older facts about the same entity or state variable. A strong mechanism claim would require labeled stale-

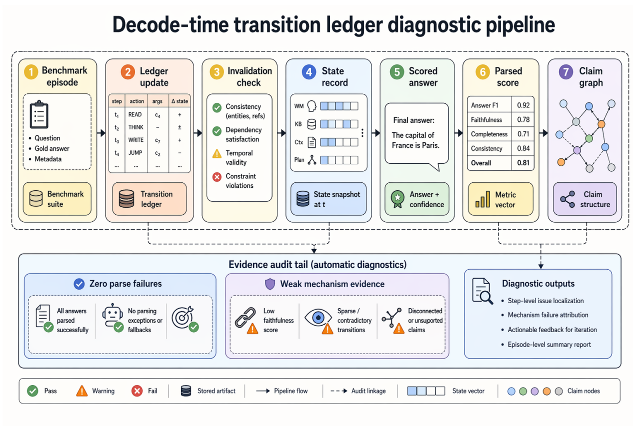

Figure 2: Method pipeline. The train-free controller produces ledger records, invalidation checks, parsed scores, and a claim graph; the zero-parse-failure claim is supported, while mechanism evidence remains weak.

fact errors and evidence that invalidation reduces exactly those errors. The current rows only show that reverify beats the no-invalidation variant on MuMA-ToM and not on the other two families. The method section therefore defines invalidation as an inspectable operation without claiming it is the measured cause of every improvement.

The claim graph is the final component of the method. It takes the result table, paired tests, and evidence gaps as inputs and assigns paper-facing claim statuses. The graph is deliberately conservative. It supports matrix coverage and zero parse failures, rejects broad wins, and keeps mechanism/resource claims weak when the required supporting evidence is absent. This is the mechanism that turns raw rows into a reviewable negative/mixed paper draft.

Experimental Setup

The completed matrix uses three benchmark families: FANToM, MMToM-QA, and MuMA-ToM. Each family contributes 67 matched task rows for each of eleven conditions, producing 33 family-condition cells and 2,211 scored rows. The conditions are direct answering, chain-of-thought, ReAct-style scratchpad, StateAct-like state prompting, summarize-then-answer, reformatted events, FactTrack-like prompting, ledger + reverify, ledger without invalidation, model ledger, and oracle ledger. All conditions use the same recorded "gpt-5-mini" OpenAI-compatible backend, and all comparisons here are within the same executed matrix.

The primary metric is mean task score with bootstrap confidence intervals over task rows. Paired comparisons use matched family and task identifiers, bootstrap confidence intervals over score differences, and paired sign-flip tests. The resource footprint is retained in replay metadata, but the paper does not use it as a primary outcome.

The matrix uses matched task identifiers for paired tests wherever conditions are compared. This matters because raw family means can hide which tasks changed status under an intervention. The paired sign-flip test asks whether the score differences on matched tasks consistently favor one condition. The bootstrap intervals provide an effect-size range, while the sign-flip p-values keep the interpretation from treating small and inconsistent deltas as reliable wins.

The scoring audit records parse failures alongside the final task scores. Parse failures are a supported scientific claim because the scorer parsed all 2,211 rows. Resource measurements are retained for replay and audit metadata, but the paper does not interpret them as a primary method claim.

Family-separated reporting is fixed before interpretation. FANToM contains belief and fact questions over multi-speaker stories, MMToM-QA stresses multi-agent mental-state questions, and MuMA-ToM contributes a separate theory-of-mind construction with different answer style. Those differences make the 67-row family cells the unit of interpretation. The paired tests therefore match family and task identifiers before comparing conditions; the paper reports bootstrap intervals over paired score differences and paired sign-flip p-values, but it does not claim multi-seed significance.

Results

Table 1 is the main result. It keeps benchmark families separate because the family direction reverses. FANToM is led by the FactTrack-like condition at 64.4%, MMToM-QA by reformatted events at 88.1%, and MuMA-ToM by chain-of-thought at 89.6%. Ledger + reverify is therefore not the headline winner in any family, although it is slightly above direct on FANToM.

The FANToM pattern is the most favorable to the ledger story but still not a clean win. Ledger + reverify reaches 60.4FactTrack-like prompting reaches 64.4difference is +1.0 pp with a confidence interval spanning negative and positive values. This supports only a cautious statement: ledger + reverify is competitive on FANToM and slightly above direct in the observed mean, but the family does not establish robust superiority over strong controls.

The MMToM-QA pattern reverses the story. Reformatted events reach 88.1answering reaches 74.6ledger-minus-reformatted-events difference is -19.4 pp with p=0.0071. This is the strongest evidence that the ledger prompt is not merely unlocking a better state representation; a simpler presentation change can be much stronger on this family.

The MuMA-ToM pattern creates a second reversal. Chain-of-thought reaches 89.6direct answering reaches 82.1chain-of-thought the ledger is lower by 11.9 pp with p=0.0406. That result is why the conclusion names the ledger an audit surface rather than a replacement for reasoning traces.

Analysis and Ablations

The paired tests match the table-level interpretation. Ledger + reverify versus direct has a +1.0 pp mean difference on FANToM, -6.0 pp on MMToM-QA, and -4.5 pp on MuMA-ToM, with paired sign-flip p-values 0.7835, 0.4962, and 0.5807. Against chain-of-thought, ledger + reverify is lower

Condition	Role	Backend / metric / budget	FANToM 67	MMToM-QA 67	MuMA-ToM 67
Direct	answer-only baseline	"gpt-5-mini"; mean score; train-free prompt budget	59.4	74.6	82.1
CoT	reasoning trace baseline	same backend and metric	56.9	73.1	**89.6**
Reformatted events	presentation control	same backend and metric	61.3	**88.1**	79.1
FactTrack-like	fact-tracking prompt control	same backend and metric	**64.4**	64.2	76.1
Ledger + reverify	transition-ledger diagnostic	same backend and metric	60.4	68.7	77.6
No invalidation	ledger ablation	same backend and metric	61.3	71.6	74.6
Model ledger	extraction diagnostic	same backend and metric	57.5	70.1	83.6
Oracle ledger	supplied-state diagnostic	same backend and metric	57.7	71.6	82.1

Table 1: Main cross-benchmark matrix over 2,211 scored rows from FANToM, MMToM-QA, and MuMA-ToM. Bold values are family winners; ledger + reverify reaches 60.4%, 68.7%, and 77.6%, so no pooled ledger-win headline is reported.

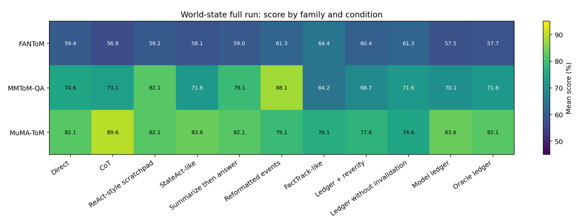

Figure 3: Data figure generated from the canonical scored rows; family winners differ across FANToM, MMToM-QA, and MuMA-ToM.

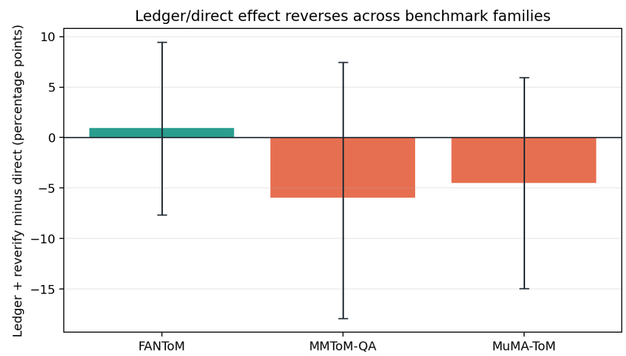

Figure 4: Data figure for ledger-minus-direct deltas: +1.0 pp on FANToM, -6.0 pp on MMToM-QA, and -4.5 pp on MuMA-ToM.

by 11.9 pp on MuMA-ToM with p=0.0406. Against reformatted events, it is lower by 19.4 pp on MMToM-QA with p=0.0071. These numbers reject the robust prompt-format-win claim.

The invalidation ablation remains weak. Ledger + reverify is below the no-invalidation variant on FANToM and MMToM-QA and above it only on MuMA-ToM. The oracle/model contrast is also weak: oracle ledgers are only slightly above model ledgers on FANToM and MMToM-QA and below on MuMA-ToM. Resource use is not analyzed as a primary claim in this paper, so the scientific interpretation rests on paired task scores, parse status, and claim support.

The prompt-format audit explains why the negative result is scientifically useful. Ledger + reverify is not compared only to direct answering; it is also compared with controls that change reasoning trace, event presentation, and state prompt structure. On MMToM-QA, reformatted events exceed ledger + reverify by 19.4 pp with a paired sign-flip p-value of 0.0071. On MuMA-ToM, chain-of-thought exceeds ledger + reverify by 11.9 pp with p=0.0406. Those two contrasts rule out the most tempting mechanism story: that writing a ledger is the uniquely effective way to expose current state. A simpler event presentation or reasoning trace can be stronger, depending on the family.

The zero-parse-failure result narrows the interpretation. Every one of the 2,211 scored rows produced a parser-readable answer, so the family reversals are not caused by one condition silently failing to emit the expected answer format. This does not prove that the scoring parser captures every semantic distinction in the task, but it removes a common implementation explanation for why a structured prompting condition might look weak.

Claim Audit and Evidence Status

Table 3 is a body result, not an internal checklist. The claim audit is necessary because the same experiment matrix contains positive, negative, and unresolved signals. Matrix coverage, family reversal, and zero parse failures are supported by the completed rows. Universal ledger improvement and robust prompt-format superiority are rejected by the same rows. The invalidation, oracle-gap, and resource-feasibility claims remain weak because the available evidence does not isolate the causal mechanism, does not make oracle ledgers a monotone upper bound, and does not analyze resource use as a primary outcome.

This accounting changes how the figures and tables should be read. The method pipeline is an audit pipeline, not a proof that invalidation is the cause of accuracy gains. The family score plot shows a structured reversal, not a noisy pooled average. The paired-delta plot shows that the

Comparison	Evidence status	FANToM	MMToM-QA	MuMA-ToM
Ledger vs direct	mixed, not robust	+1.0	-6.0	-4.5
Ledger vs no invalidation	weak mechanism evidence	-0.9	-3.0	**+3.0**
Oracle vs model ledger	no clean upper bound	+0.2	+1.5	-1.5

Table 2: Paired deltas in percentage points show weak mechanism evidence over matched tasks: invalidation helps only on MuMA-ToM, and oracle ledgers do not cleanly upper-bound model ledgers.

Claim	Status	Paper wording
Matrix coverage	**Supported**	2,211 rows; 33 family-condition cells
Universal ledger win	Rejected	no pooled win claim
Family reversal	**Supported**	report families separately
Prompt-format win	Rejected	controls beat ledger in places
Invalidation mechanism	Weak	hypothesis only; positive only on MuMA-ToM
Oracle extraction gap	Weak	no clean upper bound
Resource feasibility	Weak	not analyzed as a primary claim
Parse failures	**Supported**	zero parse failures
Error taxonomy	Missing	diagnostic triage only

Table 3: Claim audit with three supported claims, two rejected claims, three weak claims, and one missing claim, preventing the mixed ledger results from being overstated.

ledger can be competitive with direct answering on one family while still losing to presentation and reasoning controls elsewhere. The paper therefore treats the ledger as an instrument for locating unsupported claims.

The sampled error taxonomy is used only for triage. It contains 36 deterministic incorrect-row samples from direct, chain-of-thought, reformatted-events, and ledger + reverify conditions. The categories include 18 state-transition mechanism cases, 7 entity-fact normalization cases, 5 stale-fact or perspective cases, 4 ledger-mechanism-unisolated cases, and 2 invalidation-ablation regressions. Those counts are enough to motivate future labeling, but they are not enough to support an audited stale-fact mechanism claim.

Discussion

The result changes the interpretation of transition ledgers. A ledger can be valuable even when it is not an accuracy-improving intervention, because it creates a structured trace that can be audited against current-state questions. In this matrix, that trace is most useful as a way to reject overly broad claims. The same rows that make the ledger idea plausible on FANToM also show that a formatting control dominates MMToM-QA and that chain-of-thought dominates MuMA-ToM.

This is why the draft avoids a single pooled score. A pooled score would blend a small positive FANToM delta with larger negative deltas elsewhere and would invite readers to treat the average as a property of the method. The family-separated table instead makes the research object visible: state-tracking interventions interact with benchmark family, prompt format, and answer style. That interaction is the finding.

The evidence also shows why an explicit claim graph belongs in the paper rather than only in internal logs. Without the graph, the method diagram and the FANToM delta could be read as a success story. With the graph, the reader sees that the universal-win claim and the robust prompt-format claim are rejected, while mechanism and resource claims are weak. This form of claim accounting is especially important for agent and memory papers, where plausible mechanisms can sound stronger than their measured support.

There is a practical design lesson for future ledgers. The current controller asks the model to externalize state, invalidate stale facts, and answer from the record in one prompt family. Future variants should separate those operations more cleanly, so that extraction quality, invalidation quality, and final answer use can be measured independently. The oracle/model contrast in this draft is a first step, but the non-monotone results show that it is not yet a sufficient decomposition.

Finally, the negative controls point to a simpler explanation that must be ruled out in later work: some benefits attributed to state records may instead come from making events easier to read. The reformatted-events condition is a strong baseline precisely because it changes presentation without adding a ledger. Any future positive ledger claim should beat that control on the same matched task rows before claiming a state-tracking mechanism.

The supported zero-parse-failure claim is small but important for interpretation. When a benchmark matrix contains many prompting formats, one concern is that a condition may look weak because the answer parser failed more often. That did not happen here. The parsed-row count covers the full matrix, so the family reversals should be treated as measured behavior rather than missing-data noise.

The matrix also suggests how to design stronger future ablations. A useful next test would hold the rendered event format fixed while independently toggling the presence of a

Sampled category	Count	Claim effect
State-transition mechanism	**18**	State-update diagnostic, not proof that ledgers repair the update.
Entity-fact normalization	7	Limits mechanism claims because some losses are answer-rendering errors.
Stale-fact or perspective	5	Motivates future labels; current stale-fact claim remains missing.
Ledger mechanism unisolated	4	Shows why prompt-format controls are required.
Invalidation regression	2	Keeps the invalidation mechanism weak.

Table 4: Sampled failure taxonomy over 36 deterministic incorrect-row samples. The largest category is state-transition mechanism (18/36), but the sample is diagnostic triage rather than an audited stale-fact label set.

ledger, the invalidation rule, and the final reverification step. That design would make it harder for a prompt-format gain to masquerade as a state-tracking gain. The present draft cannot make that decomposition because the completed conditions do not isolate all three factors at once.

A second next test would attach human-audited stale-fact labels to matched task rows. The sampled taxonomy indicates possible categories, but only audited labels could answer whether invalidation reduces stale-fact errors rather than shifting errors to extraction or answer rendering. Until that label set exists, the mechanism claim remains a hypothesis.

These caveats do not make the result uninformative. They make the result more reviewable. The draft states exactly which rows exist, which family directions reverse, which controls defeat the ledger, and which explanations still lack evidence. For a diagnostic paper about world-state drift, that calibrated negative evidence is the main contribution.

Failure Cases and Scope

The analysis artifacts contain a sampled taxonomy of 36 rows, with categories such as state-transition mechanism, stale-fact or perspective errors, entity-fact normalization, and ledger-mechanism ambiguity. That sample is useful for triage, but the claim graph marks the audited error-taxonomy claim as missing. This draft therefore avoids example-level claims that would require a fully labeled stale-fact audit.

Table 4 gives the paper-facing version of the failure sample. The dominant category, state-transition mechanism, confirms that many sampled failures arise in rows where the model must track a changing state rather than answer from a static fact. That observation is consistent with the paper's motivation, but it does not rescue the ledger result. A diagnostic category says where the problem appears, not which prompt component fixes it. Because ledger + reverify loses to event reformatting on MMToM-QA and to chain-of-thought on MuMA-ToM, the taxonomy is read as an explanation of why the task is hard, not as proof of a successful mechanism.

The entity-normalization and stale-perspective categories also constrain the story. Entity-fact normalization cases remind us that some row-level losses come from answer rendering or entity reference, not necessarily from a stale state. Stale-fact or perspective cases are closer to the intended mechanism, but there are only five in the deterministic sample and they are not human-audited labels. The correct paper claim is therefore narrow: the sample identifies plausible directions for a future labeled audit, while the current matrix supports only the family-level score and parse-status claims.

The two explicitly ledger-facing categories explain the weak ablations. Ledger-mechanism-unisolated rows show that a ledger prompt changes several variables at once: it asks for state records, changes the output format, and increases the amount of intermediate text available to the model. Invalidation regression rows show the opposite danger: adding the invalidation instruction does not guarantee an improvement. Together these categories justify the paper's conservative conclusion. The ledger is a useful audit surface because it creates interpretable intermediate records, but the completed evidence does not identify invalidation, extraction, or reverification as the reliable cause of better answers.

Representative failures follow the same pattern. FANToM samples include stale-fact or perspective rows under direct, chain-of-thought, reformatted events, and ledger + reverify, so a state-tracking prompt alone does not remove the perspective error. MMToM-QA samples are dominated by state-transition mechanism rows, yet the event-reformatting control is the family winner. MuMA-ToM contains the only positive no-invalidation delta for ledger + reverify, but the best family score still comes from chain-of-thought. These examples connect the sampled categories back to the claim audit: the failure analysis motivates the diagnostic, while the paired scores decide what the paper can claim.

The first representative pattern is perspective leakage. In these rows, the episode contains information that is true for one participant or time slice but not available to the queried perspective. A ledger record can list the relevant events and still fail if the final answer conditions on the wrong observer. This pattern explains why final-answer scoring and ledger inspection are both needed: the intermediate state may look structured, but the answer can still bind that state to the wrong perspective.

The second pattern is presentation sensitivity. MMToM-QA is the clearest case: the reformatted-events control leads the family even though it does not add a transition ledger. That result suggests that the model sometimes benefits more from a clearer event surface than from an explicit state record. It also explains why the paper treats event reformatting as a serious baseline rather than a trivial control. If a ledger loses to a presentation-only control, the observed benefit cannot be attributed to state tracking without stronger isolation.

The third pattern is answer normalization. Several sampled FANToM rows contain high-overlap but imperfect free-form answers. These rows matter because they can make a condition look worse even when the model tracked much of the relevant state. They do not invalidate the family reversal, since the parser succeeds on all rows and the comparisons are matched, but they do caution against turning a row-level error category into a mechanism claim. The paper therefore reports mean task score and paired deltas, then uses the taxonomy only to interpret what future annotation should separate.

These patterns also explain why the oracle/model contrast remains unresolved. Supplying a cleaner state record removes one possible extraction error, but it does not guarantee that the answer renderer selects the right entity, the right observer, or the current rather than stale fact. The non-monotone oracle/model results are therefore expected under this taxonomy. A future upper-bound study would need to score extraction quality, invalidation quality, and answer use separately instead of treating an oracle row as a single causal intervention.

Finally, the taxonomy clarifies what evidence would change the conclusion. A strong invalidation claim would require audited rows showing that stale-fact errors decrease when invalidation is present while presentation and answer-use factors are held fixed. A strong extraction claim would require oracle ledgers to dominate model ledgers under the same answer policy. A strong efficiency claim would require resource use to be specified as a primary outcome. None of those conditions holds in the completed matrix, so the paper's contribution is the diagnostic accounting rather than a stronger mechanism result.

The scope is also bounded by the train-free protocol. Because no model weights are adapted, the method can expose a failure mode without guaranteeing that it repairs the failure. Because the benchmark families differ in construction and answer style, a pooled average would hide exactly the result this diagnostic is meant to surface.

The broader impact is therefore mostly about evaluation hygiene. Transition ledgers can make agent behavior look more systematic because they expose a structured record, but this paper shows why such records should not be equated with better task performance. A deployment-facing evaluation that reports only the presence of memory, state, or trace structure can overstate competence when the final-answer matrix is mixed. The safer use is diagnostic: report the intermediate state, keep matched controls, and state when the mechanism remains unproven.

The paper also avoids a common failure mode in negative results. It does not reinterpret a mixed score table as a hidden success by pooling families or by selecting the one favorable slice. Instead, it treats the negative controls as part of the result. That choice is important for state-tracking research because formatting, answer style, and perspective can all interact with apparent memory or ledger benefits. Future work should therefore keep the same discipline: separate representation from presentation, use matched tasks, and promote mechanism claims only after the relevant error labels are audited.

This protocol implication is concrete. A future ledger study should report at least three matched views for each family: a direct answer view, a presentation-control view that changes event readability without adding state records, and a ledger view that exposes state updates. If the ledger improves only over direct answering, the claim should be about diagnostic visibility unless it also beats the presentation control. If it beats the presentation control in one family but not another, the paper should report the family split rather than a pooled average. This draft follows that rule because the observed winners differ across FANToM, MMToM-QA, and MuMA-ToM.

A second implication is that parser reliability and mechanism reliability are different checks. Zero parse failures support the claim that missing or unreadable answers do not explain the reversal. They do not show that the ledger extracted the right transition, invalidated the right stale fact, or used the record correctly at answer time. That separation is useful for review: it keeps a clean scoring pipeline from being mistaken for a validated cognitive mechanism.

A third implication concerns oracle conditions. Oracle ledgers are tempting to describe as upper bounds, but this matrix does not justify that language. If an oracle state record still has to be converted into an answer, then extraction is not the only possible bottleneck. A credible upper-bound experiment would need to hold answer rendering fixed or score state extraction directly. Until then, oracle/model comparisons should be described as diagnostic contrasts rather than as proof that extraction is or is not the limiting factor.

Conclusion

Decode-time transition ledgers provide a useful diagnostic record, but the completed matrix does not support a universal ledger-win story. The evidence supports complete coverage, family-specific reversals, and zero parse failures; it rejects universal and prompt-format robustness claims; and it leaves mechanism, oracle-gap, resource, and audited failure-taxonomy claims weak or missing. The practical lesson is to treat transition ledgers as an audit surface for world-state drift, not as a guaranteed accuracy intervention.

The broader methodological lesson is that explicit state records should be evaluated with controls that separate representation, presentation, and answer use. In this matrix, a ledger can make the intermediate state easier to inspect while still losing to event reformatting or chain-of-thought on final-answer score. Reporting that tension is the point of the diagnostic: it tells future work which mechanism claims require new labels or cleaner ablations, and it prevents a plausible state-tracking story from being promoted beyond the evidence.

For that reason, the negative rows are not secondary cleanup. They are the evidence that makes the paper useful. The paired task design shows where the ledger moves with direct answering, where formatting dominates, and where reasoning traces remain stronger. The claim graph then turns those observations into explicit support levels. That combination is the reusable contribution: a paper can study world-state drift honestly even when the proposed controller is not the best final-answer method.

The resulting evaluation recipe is simple enough to reuse. Start with matched tasks across benchmark families, include controls that change reasoning trace and event presentation, keep family-level scores visible, and write down which claims the rows support, reject, weaken, or leave missing. Then use any ledger or memory trace as an object of diagnosis rather than as self-validating evidence. Under that recipe, a mixed result is not a failed paper shape. It is a map of where the proposed state representation interacts with benchmark design, answer style, and the model's existing reasoning behavior.

The same recipe also makes future positive results more credible. If a later ledger variant wins after the presentation and reasoning controls are fixed, the claim will be stronger because the obvious confounds have already been tested. If it still fails on some families, the family split will identify where the state representation needs a different parser, verifier, or task interface. Either outcome is more informative than a single pooled score. For world-state drift, that is the standard this paper argues for: diagnose the transition, verify the answer, and keep the claim no stronger than the matched evidence.

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Reproducibility Checklist

This draft uses public benchmark-family rows assembled into a three-family, eleven-condition diagnostic matrix. The paper-facing replay interface is: select the same family split, run the eleven condition prompts with the same hosted backend and deterministic scoring parser, regenerate the result table, claim graph, data figures, and paired tests, then compare the claim statuses. The evaluated runs are train-free and use one seed-equivalent deterministic prompting configuration per condition. Replay requires the same benchmark split, condition prompts, evaluated backend, deterministic answer parser, metric definitions, paired-test policy, and generated figures and tables. No paper-facing claim depends on private execution settings or manual metric correction.

Appendix: Evidence Mapping

The appendix records how claims are constrained. The main table and figures are derived from the completed scored-row analysis. Bibliography entries have scholarly metadata, and every result claim is either supported, rejected, weak, or missing according to the evidence available in the completed matrix.

Additional result detail: the family-specific best conditions are FactTrack-like on FANToM, reformatted events on MMToM-QA, and chain-of-thought on MuMA-ToM. Those winners are intentionally not collapsed into a pooled score. The condition set also includes direct answering, ReAct-style scratchpad, StateAct-like prompting, summary-before-answer, ledger without invalidation, model ledger, and oracle ledger. The analysis uses those controls to distinguish ledger structure from prompt-format changes.

Additional claim detail: the supported claims are matrix completion, family reversals, and zero parse failures. The rejected claims are universal ledger improvement and robust prompt-format superiority. The weak claims are invalidation mechanism, oracle/model extraction gap, and resource feasibility. The missing claim is the audited error taxonomy. These labels are part of the paper's scientific result and should remain visible to reviewers.

Additional figure detail: Figure 1 and Figure 2 are conceptual visuals for the claim framing and method pipeline. Figure 3 and Figure 4 are data plots generated from scored rows.